EfficientNet-DenseNet Network with Bidirectional Cross-Attention and parallel GradCAM mechanism (ED-XAGN) for Brain Tumor Diagnosis in MRIs

Introduction

Brain tumors represent one of the most aggressive and clinically challenging neurological disorders, with their early and precise diagnosis being paramount for determining appropriate treatment strategies and improving patient outcomes. As the primary diagnostic modality for brain tumor detection, Magnetic Resonance Imaging (MRI) offers exceptional soft tissue contrast and non-invasive visualization capabilities that are indispensable in clinical neurology. However, the manual interpretation of MRI scans remains fraught with significant challenges that compromise diagnostic reliability, including substantial inter-observer variability among radiologists, the inherent subjectivity of visual assessment, and the growing problem of diagnostic fatigue in high-volume clinical settings. These limitations become particularly pronounced when attempting to differentiate between common tumor types such as gliomas (the most aggressive primary brain tumors), meningiomas (typically benign but potentially problematic tumors), and pituitary tumors (which present unique diagnostic challenges due to their anatomical location).

Traditionally, brain tumors have been detected using imaging techniques such as magnetic resonance imaging (MRI) and computer tomography (CT) scans. These methods rely on the visual assessment of scans by trained radiologists to identify abnormalities. However, Conventional techniques for detecting brain tumors have limitations such as variability in human interpretation and the difficulty of detecting subtle early signs of tumors. Due to these challenges machine learning (ML), especially transfer learning (TL) has been employed in the early detection and classification of brain tumors.
Computer-Aided Diagnosis (CAD) systems leveraging deep learning and transfer learning paradigms have been deployed. Although standard Convolutional Neural Networks (CNNs) demonstrate strong capabilities in local spatial feature extraction, traditional architectures encounter operational limitations when applied to neuro-oncologic imaging:

1. *Sensitivity to Irrelevant Spatial Artifacts*: Standard CNN backbones process background cranial bone, soft tissue, and acquisition noise uniformly alongside parenchymal pathology, degrading feature representation purity.
2. *Receptive Field and Contextual Bottlenecks*: Standard localized convolutional

kernels struggle to capture broad multi-scale spatial dependencies and context across complex, heterogeneous tumor microenvironments.
3. *Loss of Spatial Coordinate Geometry in Feature Fusion*: Naive feature fusion methods collapse 2D spatial feature maps via premature Global Average Pooling (GAP) prior to vector concatenation or element-wise addition, destroying 2D coordinate layouts and restricting explainability.
4. *Predictive Opacity and Lack of Reliability Quantification*: Conventional point-estimate deep classifiers do not quantify diagnostic uncertainty, offering clinicians no indication of when a model prediction is ambiguous or unconfident.

Literature Survey

Baseline Model: An attention-fused architecture for brain tumor diagnosis

Arash Hekmat, Zuping Zhang, Saif Ur Rehman Khan, Ifza Shad, Omair Bilal, An attention-fused architecture for brain tumor diagnosis, Biomedical Signal Processing and Control, Volume 101, 2025, 107221, ISSN 1746-8094, https://doi.org/10.1016/j.bspc.2024.107221.

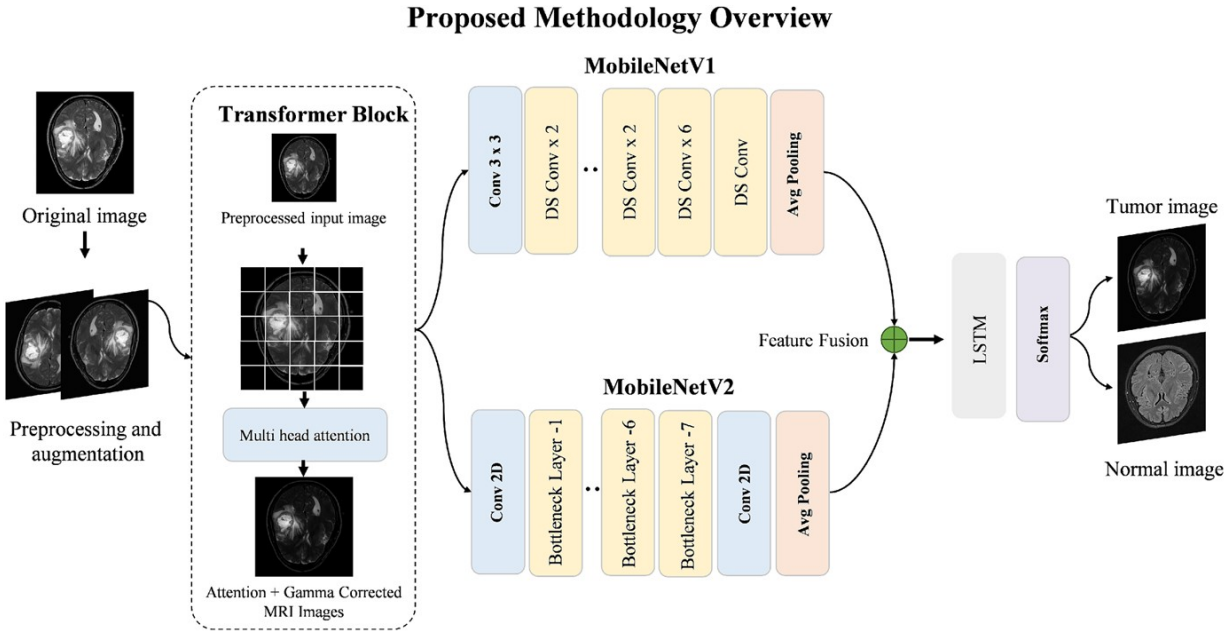
The paper introduces a novel method, the Attention-Fused MobileNet-LSTM model, which aims to improve the classification accuracy on Brain Tumor MRIs. By integrating features from two well recognized pre-trained convolutional neural network (CNN) models named as MobileNetV1 and MobileNetV2, the paper has developed a reliable model. Their approach addresses challenges in CNN training, including high processing demands, difficulty in discerning intricate details in images, and diminished efficacy due to noise or irrelevant data. They employ a transformer aid in feature extraction, while gamma correction before features fusion tackles long-range relationship capture between successive features. The proposed model employs Long Short-Term Memory (LSTM) recurrent neural networks (RNN) for direct class label prediction and understanding the intermediate features from feature fusion.
They leverage a publicly available brain tumor dataset comprising 3,000 MRI images, on which they conduct augmentation procedures and standardization to meet the CNN's image resizing prerequisites, thereby enhancing dataset diversity and classification task performance. With a consistent accuracy of 98.66%, their proposed model showcases significant potential as a reliable tool for precise classification of normal and malignant brain MRIs in the medical domain. Additionally, they employ Grad-CAM to visualize the key regions prioritized by their proposed model, offering insights into their decision-making processes.

Operational Workflow

1. *Transformer Attention Preprocessing*: Raw MRI frames are passed through a Vision Transformer (ViTL16) self-attention encoder to suppress non-brain anatomical structures and background noise.

2. *Photometric Luminance (Gamma) Calibration*: The attention-filtered representation undergoes gamma correction to calibrate contrast across dynamic ranges.
3. *Dual Backbone Extraction*: The contrast-calibrated image is fed simultaneously into two parallel pre-trained backbones: MobileNetV1 (86 layers, 3.23 million parameters) and MobileNetV2 (154 layers, 2.26 million parameters).
4. *Vector Flattening and Concatenation*: Output feature maps are flattened via Global Average Pooling (GAP) and merged in a 1D vector concatenation layer, yielding a 2,304-dimensional vector.
5. *Recurrent Sequence Modeling*: The concatenated vector is routed into a Long Short-Term Memory (LSTM) recurrent network to capture inter-feature sequential dependencies prior to SoftMax probability projection.

Architectural Diagram



Architectural Limitations

- *Latent Feature Opacity and Loss of Interpretability*: Unrolling non-sequential spatial feature vectors across artificial temporal steps in an LSTM obscures feature attributions. Because the LSTM updates internal cell states (C_t) and hidden states (h_t) through non-linear activation gates (i_t, f_t, o_t), individual feature contributions from MobileNetV1 versus MobileNetV2 become entangled. This limits the ability of clinicians to inspect intermediate spatial maps or verify prediction logic.
- *Architectural Redundancy and also Low Information Gain*: Both MobileNetV1 and MobileNetV2 belong to the same lightweight architectural family, relying heavily on depthwise separable convolutions and inverted residual bottleneck blocks. Because their feature extraction mechanisms are structurally similar,

their feature representations overlap significantly. Fusing two similar models provides lower information gain compared to combining complementary, functionally specialized backbones.

- *Premature Spatial Collapse*: Applying Global Average Pooling (GAP) before feature fusion flattens 7 x 7 2D spatial feature maps into 1D vectors. This discards 2D spatial coordinates prior to cross-feature interaction, preventing the model from performing explicit spatial query-key alignment across branches.
- *Absence of Predictive Uncertainty Quantification*: The AFML network outputs single point-estimate class probabilities. It cannot measure predictive variance or clinical uncertainty, offering no mechanism to flag ambiguous scans for expert radiologist review.

Methodology

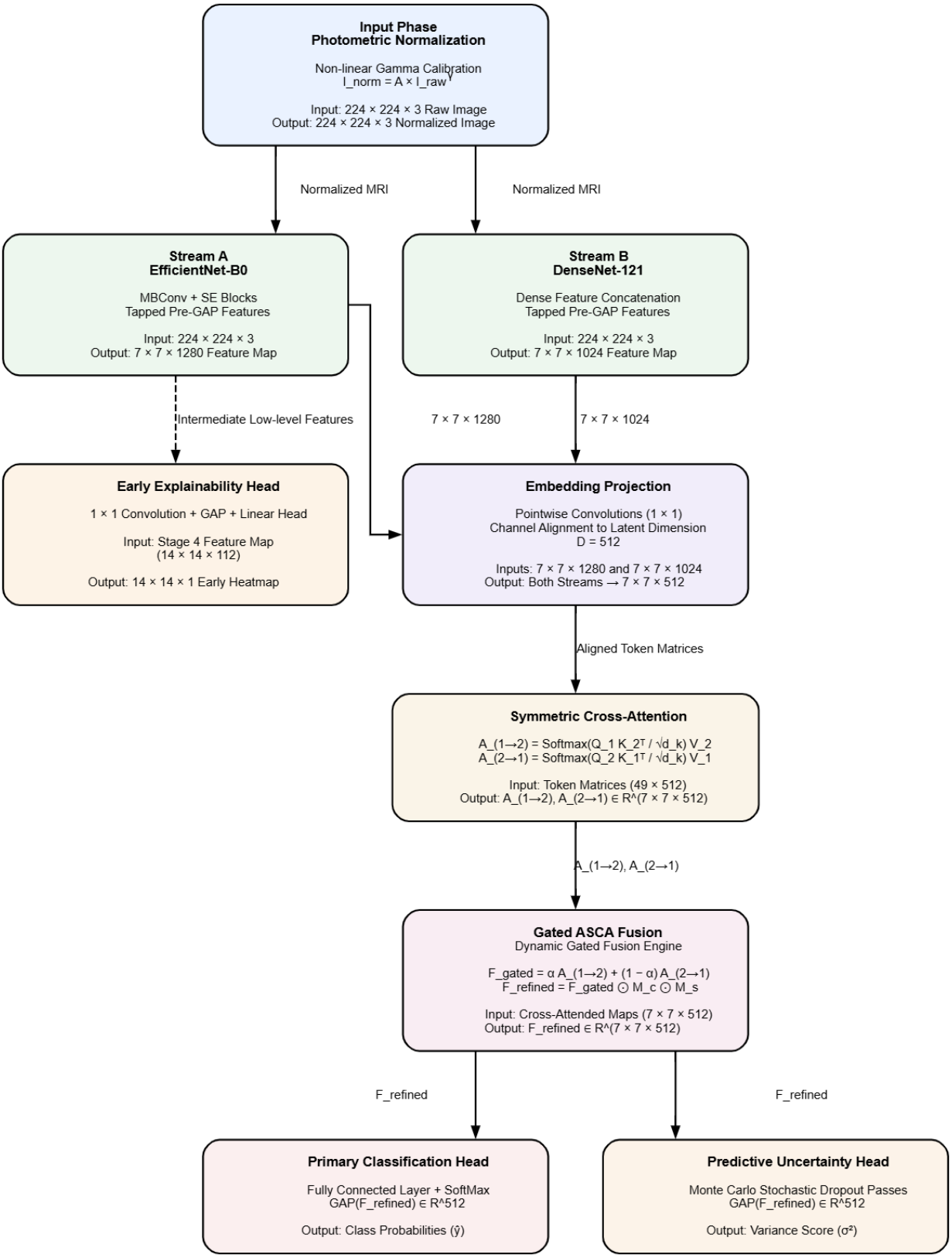
Proposed Model: EfficientNet-DenseNet Network with Bidirectional Cross-Attention and parallel GradCAM mechanism (ED-XAGN)

Rationale and Architectural Updates

1. *Backbone Replacement for Functional Specialization*: We replace MobileNetV1 and MobileNetV2 with two complementary, structurally distinct CNN backbones:
    - *EfficientNet-B0 (Local Spatial & Texture)*: Utilizes compound scaling (balancing depth, width, and resolution) with mobile inverted bottleneck (MBConv) blocks and built-in Squeeze-and-Excitation (SE) channel attention. It extracts local boundary geometry, fine tumor edge transitions, micro-calcifications, and surface textures.
    - *DenseNet-121 (Multi-Scale Semantic)*: Utilizes direct shortcut connections where every layer receives concatenated feature maps from all preceding layers. It preserves low-level, mid-level, and high-level feature maps simultaneously, capturing rich multi-scale semantic abstractions and long-range anatomical context.
2. *Removal of LSTM*: We eliminate the LSTM module entirely. Instead of treating static spatial vectors as dynamic time series, feature interaction is handled by an explicit, transparent *Bidirectional Cross-Attention Engine* that operates directly on 2D spatial feature maps.
3. *Preservation of 2D Spatial Layout:* Feature maps are tapped prior to Global Average Pooling at their spatial resolution (7×7), maintaining 2D spatial coordinates throughout cross-attention querying and gated fusion.
4. *Single-Stream Early Parallel Grad-CAM Mechanism*: We eliminate the late-stage Grad-CAM overhead and replace it entirely with a single-stream early parallel Grad-CAM head attached directly to an intermediate stage of EfficientNet-B0 (14×14 resolution). This generates a high-resolution early

Grad-CAM heatmap focused on local structural edge boundaries in parallel with the main cross-attention pipeline.

5. *Clinical Decision Support and Predictive Uncertainty*: The output stage incorporates a dual-head multi-layer perceptron (MLP) containing a primary classification head alongside an *Uncertainty Estimation Head*. Scans with high predictive variance are flagged automatically for escalation.



Architectural Breakdown

1: Dynamic Photometric (Gamma) Normalization

Raw axial MRI scans $I_{\text{raw}} \in \mathbb{R}^{224\times224\times3}$ undergo rescaling to normalized floating-point range [0.0, 1.0]. To standardize dynamic range shifts across differing hospital scanner protocols, a non-linear gamma correction transformation is applied:
$I_{\text{gamma}} = A \cdot I_{\text{raw}}^{\gamma}$
Where A = 1.0 and γ = 1.2. This transformation expands visual contrast along low-intensity tumor boundaries and peritumoral edema while preserving parenchymal details.

2: Parallel Specialized Feature Backbones

The contrast-calibrated image tensor I_{gamma} is passed simultaneously into two parallel feature extractors tapped prior to Global Average Pooling (GAP):

1. ***EfficientNet-B0 Stream***: Outputs a 2D spatial tensor $F_{\text{Eff}} \in \mathbb{R}^{7\times7\times1280}$. Its compound scaling mechanism isolates high-frequency spatial boundaries, local edge transitions, and micro-texture anomalies.
2. ***DenseNet-121 Stream***: Outputs a 2D spatial tensor $F_{\text{Dense}} \in \mathbb{R}^{7\times7\times1024}$. Its dense connectivity reuses feature maps from early, intermediate, and deep layers, capturing multi-scale anatomical representations and spatial context.

3: *Single-Stream Parallel Grad-CAM Engine (EfficientNet-B0)*

Rather than computing Grad-CAM at the end of deep cross-attention, our architecture relies exclusively on an early parallel explainability head attached directly to EfficientNet-B0:

1. ***Intermediate Spatial Feature Tap***: An intermediate forward hook taps MBConv Stage 4 of EfficientNet-B0, yielding $F_{\text{Eff_early}} \in \mathbb{R}^{14\times14\times112}$.
2. ***Bottleneck Channel Reduction (1×1 Conv)***: Channels are compressed from 112 to 64: $F_{\text{early}} = \text{ReLU}\left(\text{BatchNorm}\left(\text{Conv}_{1\times1}(F_{\text{Eff_early}})\right)\right) \in \mathbb{R}^{14\times14\times64}$
3. ***Auxiliary Classifier Head***: Global Average Pooling collapses F_{early} into a 64×1 vector passed to a linear projection layer outputting auxiliary class predictions $\hat{y}_{\text{aux}} : \hat{y}_{\text{aux}} = \text{Softmax}(W_{\text{aux}} \cdot \text{GAP}(F_{\text{early}}) + b_{\text{aux}})$
4. ***Early Grad-CAM Backprop Pass***: Visual explanations are generated by back-propagating gradients from the auxiliary class score y_{aux}^{c} into
   $F_{\text{early}} : \alpha_{\text{early},k}^{c} = \frac{1}{196}\sum_{i=1}^{14}\sum_{j=1}^{14}\frac{\partial y_{\text{aux}}^{c}}{\partial F_{\text{early},i,j}^{k}}$
   $L_{\text{Grad-CAM}}^{c} = \text{ReLU}\left(\sum_{k}\alpha_{\text{early},k}^{c}F_{\text{early}}^{k}\right) \in \mathbb{R}^{14\times14}$
5. ***Bilinear Upsampling***: $L_{\text{Grad-CAM}}^{c}$ is upsampled from 14×14 to 224×224, generating a high-resolution heatmap highlighting low-level spatial tumor boundaries.

4: Linear Projection into Shared Embedding Space

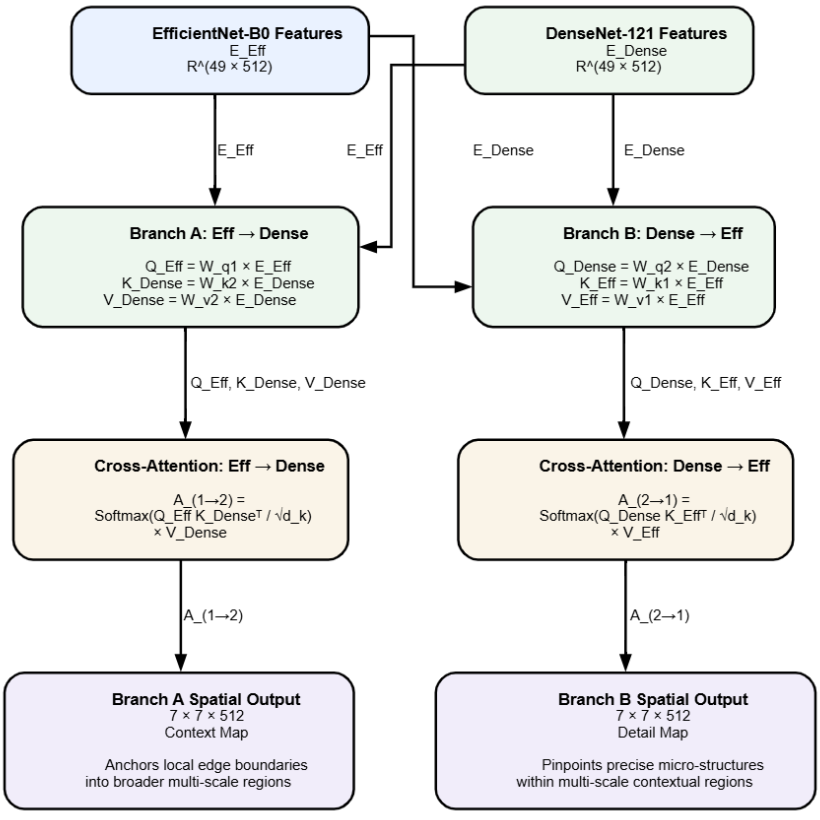
Because F_{Eff} has C_1 = 1280 channels and F_{Dense} has C_2 = 1024 channels, direct cross-covariance dot products require channel matching. Both spatial feature maps are

projected to a uniform latent channel dimension D = 512 using 1×1 pointwise convolutions:
$E_{\text{Eff}} = \text{Conv}_{1\times1}(F_{\text{Eff}}) \in \mathbb{R}^{7\times7\times512} E_{\text{Dense}} = \text{Conv}_{1\times1}(F_{\text{Dense}}) \in \mathbb{R}^{7\times7\times512}$
Flattening the spatial dimensions 7×7 yields token matrices $E_{\text{Eff}}, E_{\text{Dense}} \in \mathbb{R}^{49\times512}$, where N = 49 spatial tokens.

5: Symmetric Bidirectional Cross-Attention Engine

Rather than using single-direction attention, both branches query each other symmetrically:

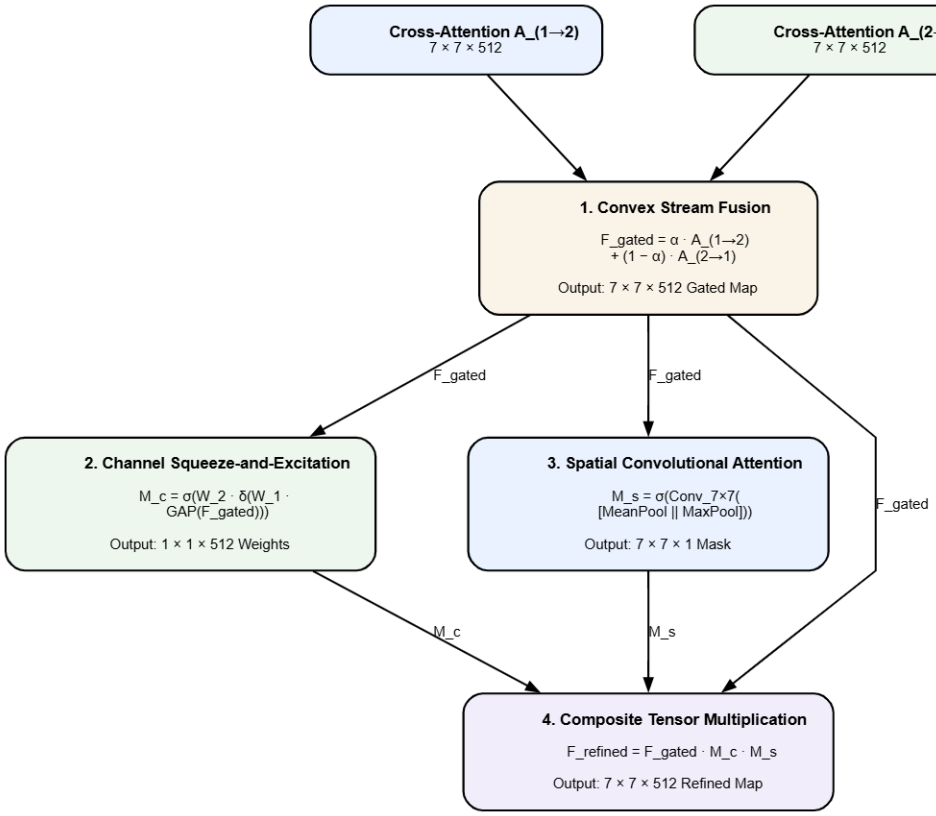
6. ***Branch 1: EfficientNet Queries DenseNet ($A_{1\to2}$)***: EfficientNet's local spatial queries $(Q_{\text{Eff}} = W_{q1}E_{\text{Eff}})$ search DenseNet's multi-scale key space $(K_{\text{Dense}} = W_{k2}E_{\text{Dense}})$ to locate where fine boundary features fit within broader anatomical context:
$A_{1\to2} = \text{Softmax}\left(\frac{Q_{\text{Eff}}K_{\text{Dense}}^T}{\sqrt{d_k}}\right) V_{\text{Dense}} \in \mathbb{R}^{49\times512} Where V_{\text{Dense}} = W_{v2}E_{\text{Dense}}$
and d_k = 64 is the projection scaling factor.
7. ***Branch 2: DenseNet Queries EfficientNet ($A_{2\to1}$)***: DenseNet's global semantic queries ($Q_{\text{Dense}} = W_{q2}E_{\text{Dense}}$) search EfficientNet's key space ($K_{\text{Eff}} = W_{k1}E_{\text{Eff}}$) to pinpoint exact micro-calcifications and edge details corresponding to pathological zones: $A_{2\to1} = \text{Softmax}\left(\frac{Q_{\text{Dense}}K_{\text{Eff}}^T}{\sqrt{d_k}}\right) V_{\text{Eff}} \in \mathbb{R}^{49\times512}$ Where $V_{\text{Eff}} = W_{v1}E_{\text{Eff}}$.
8. ***Spatial Reshaping***: Both cross-attended token representations are reshaped back into 2D spatial maps $A_{1\to2}, A_{2\to1} \in \mathbb{R}^{7\times7\times512}$.



6: Adaptive Spatial-Channel Gated Fusion Architecture (ASCA)

The cross-attended feature maps from both streams are merged using a learnable gating parameter $\alpha \in [0, 1]$:
$F_{\text{gated}} = \alpha \cdot A_{1\to2} + (1 - \alpha) \cdot A_{2\to1}$
The gating network dynamically adjusts α on an image-by-image basis depending on whether local boundary features or global multi-scale context are more informative for a given MRI scan.
The gated tensor $F_{\text{gated}} \in \mathbb{R}^{7\times7\times512}$ then passes into the *Adaptive Spatial-Channel Attention (ASCA)* module, which applies parallel channel and spatial recalibration:

1. ***Channel Attention (Squeeze-and-Excitation)***: Aggregates global spatial context via Global Average Pooling and passes it through a two-layer bottleneck MLP to weight informative feature channels while suppressing noisy channels:
   $M_c(F_{\text{gated}}) = \sigma(W_2 \cdot \delta(W_1 \cdot \text{GAP}(F_{\text{gated}}))) \in \mathbb{R}^{1\times1\times512}$ Where
   $W_1 \in \mathbb{R}^{\frac{C}{r}\times C}, W_2 \in \mathbb{R}^{C\times\frac{C}{r}}$, reduction ratio r = 16, δ denotes ReLU activation, and σ denotes Sigmoid activation.
2. ***Spatial Attention***: Applies a 7 x 7 spatial convolution across channel-pooled feature slices to construct a spatial relevance map, highlighting pathological lesion tissue relative to normal brain tissue:
   $M_s(F_{\text{gated}}) = \sigma(\text{Conv}_{7\times7}([\text{MeanPool}(F_{\text{gated}}) \parallel \text{MaxPool}(F_{\text{gated}})])) \in \mathbb{R}^{7\times7\times1}$
3. ***Composite Refinement***: $F_{\text{refined}} = F_{\text{gated}} \odot M_c(F_{\text{gated}}) \odot M_s(F_{\text{gated}})$



7: Dual Output Classifier with Predictive Uncertainty Head

The refined tensor F_{refined} is processed via Global Average Pooling into a 512-dimensional vector $v_{\text{fused}} \in \mathbb{R}^{512}$ and passed into a dual-head Multi-Layer Perceptron (MLP):

1. ***Diagnostic Classification Head***: Computes diagnostic class probabilities using SoftMax activation: $\hat{y} = \text{Softmax}(W_c \cdot v_{\text{fused}} + b_c)$
2. ***Predictive Uncertainty Head***: Estimates model prediction variance (σ^2) using Monte Carlo Dropout across T = 20 stochastic forward passes: $\text{Var}(y) = \frac{1}{T}\sum_{t=1}^{T}(\hat{y}_t - \bar{y})^2 If \text{Var}(y) > \tau_{\text{threshold}}$, the system flags the scan as *High Diagnostic Uncertainty*, routing it for manual radiologist review.

Dataset and Evaluation Metrics

The chosen dataset is the Brain-Tumor-MRI-Dataset (https://huggingface.co/datasets/BJyotibrat/Masoud-Nickparvar-Brain-Tumor-MRI-Dataset) which consists of 7,200 MRI scans. The dataset split is as follows:

CLASS NAME	AMOUNT
NO TUMOR	1,800
GLIOMA	1,800
MENINGIOMA	1,800
PITUATARY	1,800

The evaluation metrics being considered are:

Accuracy: $\frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$

Precision: $\frac{\text{TP}}{\text{TP} + \text{FP}}$

Recall: $\frac{TP}{TP + FN}$

F1 Score: $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

Hyperparameter Configurations

In order to implement the **ED-XAGN** architecture in PyTorch and Keras, the following hyperparameters are specified:

1. ***Input Image Resolution***: Resized to 224×224×3 pixels using bilinear interpolation.
2. ***Optimization Algorithm***: Adam optimizer with initial learning rate η = 0.0001 and cosine annealing decay.
3. ***Loss Function***: Composite Focal Loss combining primary and auxiliary losses: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Focal}}(y, \hat{y}_{\text{primary}}) + \lambda_{\text{aux}} \cdot \mathcal{L}_{\text{Focal}}(y, \hat{y}_{\text{aux}})$ Where $\lambda_{\text{aux}} = 0.3$ and focal scaling parameter $\gamma_{\text{focal}} = 2.0$.
4. ***Training Schedule***: batch_size = 32 with 30 training epochs.
5. ***Cross-Attention Multi-Head Setting***: H = 8 attention heads ($d_k = 64$).
6. ***Auxiliary Bottleneck Channel Count***: $C_{\text{aux}} = 64$ channels.
7. ***Uncertainty Pass Parameters***: T = 20 Monte Carlo stochastic dropout forward passes with dropout rate p = 0.3.

Conclusion

The upgraded ***EfficientNet-DenseNet Network with Bidirectional Cross-Attention and parallel GradCAM mechanism (ED-XAGN)*** directly addresses the architectural limitations of the older AFML base model. By replacing MobileNetV1/V2 with EfficientNet-B0 (local spatial specialist) and DenseNet-121 (multi-scale semantic specialist), the network maximizes feature diversity. Removing the LSTM module in favor of ***symmetric bidirectional cross-attention*** and ***ASCA gated fusion*** preserves 2D spatial layouts, enhances classification accuracy, and produces clear, artifact-free Grad-CAM visual heatmaps. Finally, the integrated ***predictive uncertainty head*** provides clinicians with a confidence metric, establishing a reliable framework for brain tumor diagnosis.